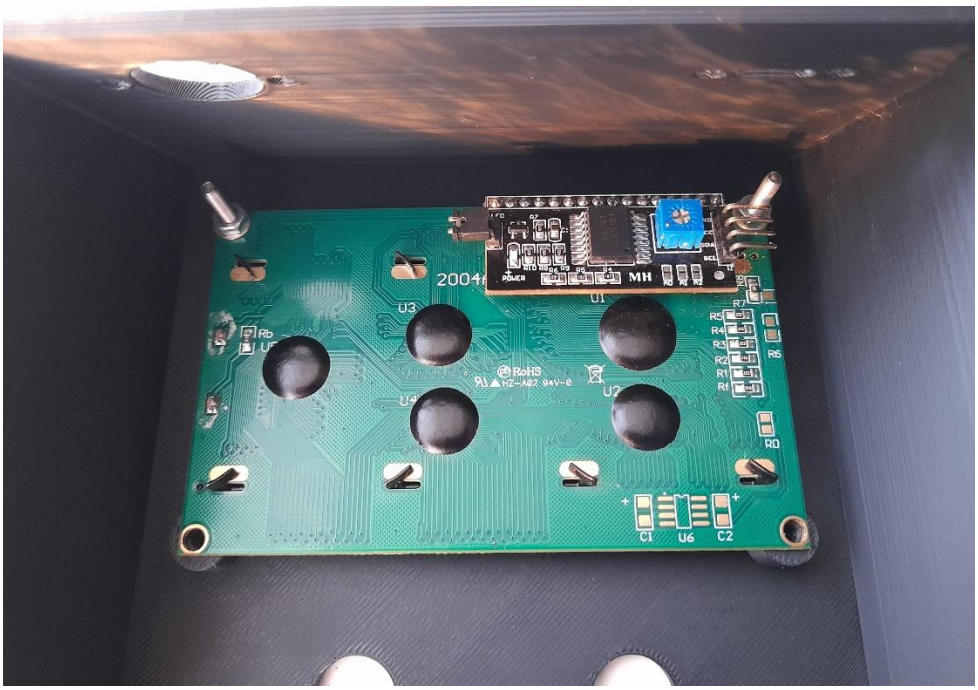
GitHub repository where all documentation & code can be found. billbill100 (github.com)

Arduino 3d print case and Sensor & Laser Enclosures V1.0 03/03/2026

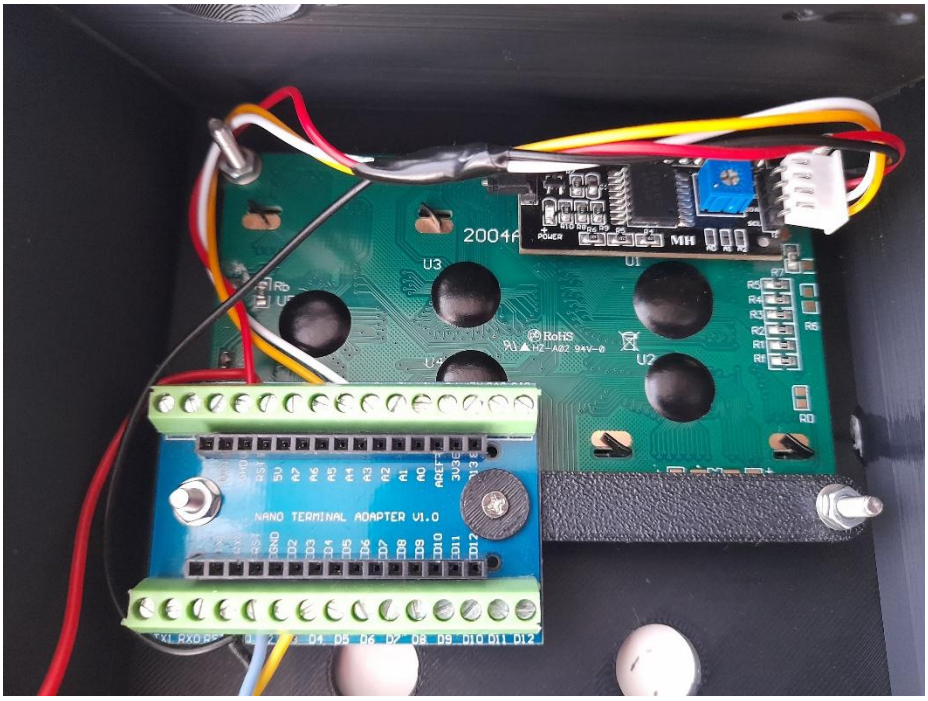
To print the case.
There are four parts to print.
1) The main case (one part front & sides).
2) Case bottom.
3) Mounting plate for screw-terminal board.
4) washer

To Assemble. Two M3 X 25mm bolts go through the two top holes in the front of the case.
The LCD is then put inside the box, with the bolts going through the top two holes and secured with washers and nuts. Do not fully tighten.

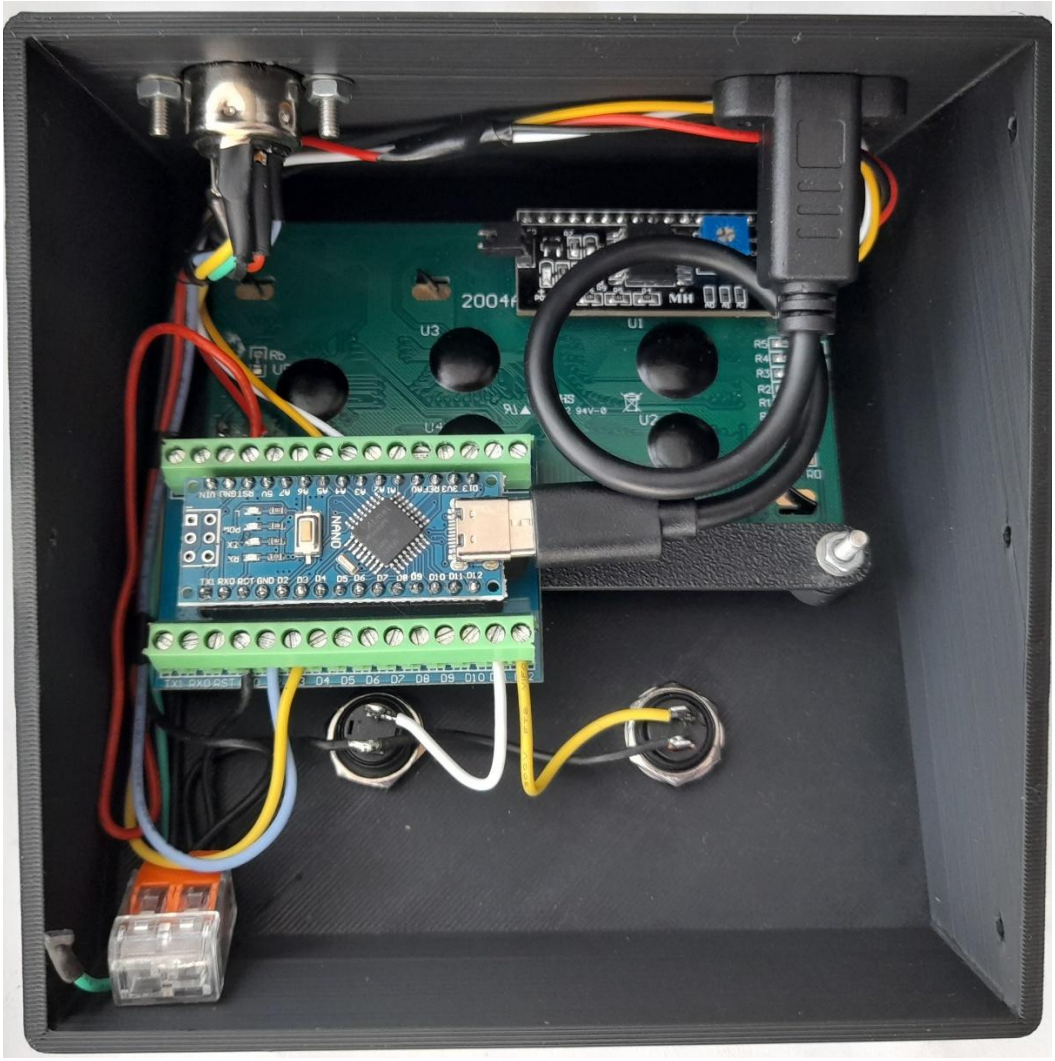


Two more bolts and board holder added. Note the small hole is slightly offset to the left.

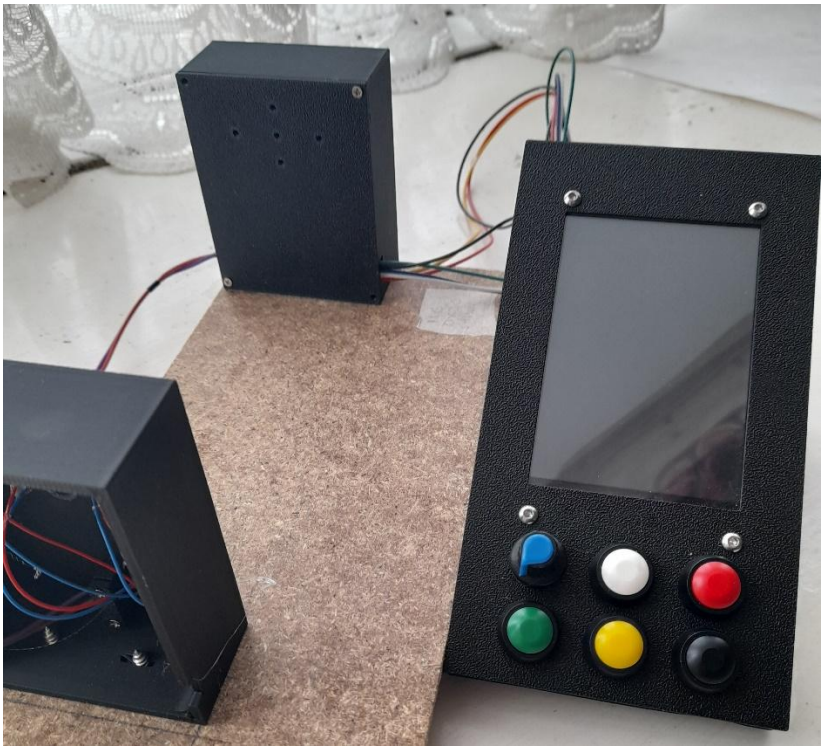
The screw-terminal board is added, with washers and nuts put on the remaining two bolts. All nuts & bolts can be tightened.
The 3d printed washer and a 2mm screw is used though the second hole of the expansion board.



Sockets & buttons are added and the wiring completed.



To print & assemble the sensor enclosure.

Please refer to the ESP32 'Print a case sensor & Laser enclosure' document, found in the ESP32 folder on Github for details of these parts. They are identical other than the Arduino using two, rather than three sensors & Lasers.

Photos showing the sensor & Laser enclosure, with the ESP32 shutter tester case.